

Global Fertility Rates Time Series Analysis Case Study Rubric

DS 4002 | Spring 2025

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Individual Assignment

Submission Format: Upload GitHub repository link to Canvas

Why am I doing this?

The goal of this assignment is to utilize your data science skills to analyze real-world data. To analyze the data you will use two analysis methods: ARIMAX and Two Way Fixed Effects (TWFE). Imagine that you are working for the UN to help predict population growth across the globe. This project will allow you to learn and apply meaningful analysis to real-world scenarios and data.

What am I going to do?

In this case study, you will explore time series analysis by investigating the influence levels of various factors on a country's fertility rate. Your goal is to determine which factor has the most predictive power on fertility rates using TWFE. Once you have determined this, your next goal should be to take your analysis and use ARIMAX to predict the fertility rate for the future. When you have completed your ARIMAX model you will compare your predictions to the predictions made by the UN using RMSE and MAE.

1. [Access the GitHub Repository](#): Access the repo and familiarize yourself with its format and contents. The repo will guide you through the project.
2. Gather and Prepare the Data: In the DATA folder obtain the dataset for each variable (and the UN predictions) and using Python clean and combine them into one cohesive csv. Some datasets do not include certain countries (of the 15 the case study is investigating) or have missing values for certain years, so be sure to account for that.
3. Perform TWFE: Create a regression model using data, spanning 35+ years, with fertility rate as the dependent variable and GDP per capita, women's education attainment, life expectancy, infant mortality rate, and homicide rate as predictors. This TWFE model will allow us to determine the strength and direction of each variable's impact on fertility rates. Data processing includes handling missing values and checking for multicollinearity among predictors.
4. Perform ARIMAX: ARIMAX stands for AutoRegressive Integrated Moving Average with Exogenous Variables. The ARIMAX model will leverage time-dependent patterns and is useful as it takes into account influencing factors, not just past fertility rates, to make a prediction.

5. Analyze and Visualize the TWFE and ARIMAX Results: The accuracy of the ARIMAX results will be evaluated using RMSE and MAE compared to the UN predictions (DATA folder). The errors for TWFE will come from printing the model's summary.
6. Deliver Insights: Summarize your findings in a clear, concise report with visualizations and results from both models. Offer recommendations on how you might improve your model in the future.

Your final deliverables should include:

- GitHub repository with all materials listed below included:
- README.md: A quick overview of the project, containing steps to reproduce the results.
- LICENSE.md: File explaining the terms under which your repo can be used and cited.
- RESOURCES folder: Folder containing all source code and materials initially available.
- SCRIPTS folder: Folder containing all scripts written over the course of the project with well commented code.
- DATA folder: Folder containing all of the raw data used and a final dataset (if needed).
- Final Report pdf: A clear, concise report that states your findings. Completed with visualizations and insights from your analysis. On a new page, there is a list of all references (sources, articles, datasets, and libraries) that were used during the project.

Tips for Success:

- Research the models: For both TWFE and ARIMAX, take a look at the resources as they are complex models. Understanding one-hot encoding is critical for the TWFE!
- Spend time with the data: There are multiple datasets being used here, so make sure you understand what each dataset is indicating and how it is relevant.
- Stay organized: Keep your code and files organized and conveniently named so that it is easy for others to help you and for you to understand your previous work.
- Ask for help: Don't hesitate to ask for help from peers, the TA, or the professor—this project has a lot of moving parts and collaboration and feedback is encouraged.

How will I know I have succeeded?

If all of the requirements in the rubric below are met, the analysis can be deemed successful.

Formatting	● GitHub repo link (submitted via Canvas) ○ README.md ○ LICENSE.md (use MIT as default) ○ RESOURCES folder ○ SCRIPTS folder ○ DATA folder ○ Final Report pdf
README.md	Concise summary regarding what you've produced for this case study.

	Contains a section regarding how to reproduce the results of the project. Serves as a way to inform people regarding your project. Should be organized with sections and headers as needed (using markdown).
LICENSE.md	This file explains to a visitor the terms under which they may use and cite your repository. Select an appropriate license from the GitHub options list on repository creation(MIT license is the default).
RESOURCES folder	This folder contains all source code and resources provided at the beginning of the project.
SCRIPTS folder	This folder contains all the code written for your project. All scripts must be named in a uniform and concise manner so that the folder is easy to follow. If possible, name the scripts in a manner so that it is clear the order they should be run in. All scripts MUST be COMMENTED throughout, and contain a commented explanation at the beginning of what they are going to accomplish.
DATA folder	This folder contains all of the raw datasets for this project and the final dataset (if it is different from the raw files). All the non-raw datasets from this folder should be able to be produced from code in the SCRIPTS folder, and all data used in the SCRIPTS folders' files should be present in the DATA folder. If your data does not fit in GitHub, upload a pdf explaining the process to obtain the dataset(s). All datasets should be named uniformly and in a concise manner so the folder is easy to manage.
Final Report pdf	A pdf file that compiles all of the generated visualizations, EDA plots, findings and error analysis of the methods (TWFE, ARIMAX with the MAE and RMSE compared to the UN). At the end there is a list of all references used in the project (articles, datasets, libraries) in IEEE documentation style .